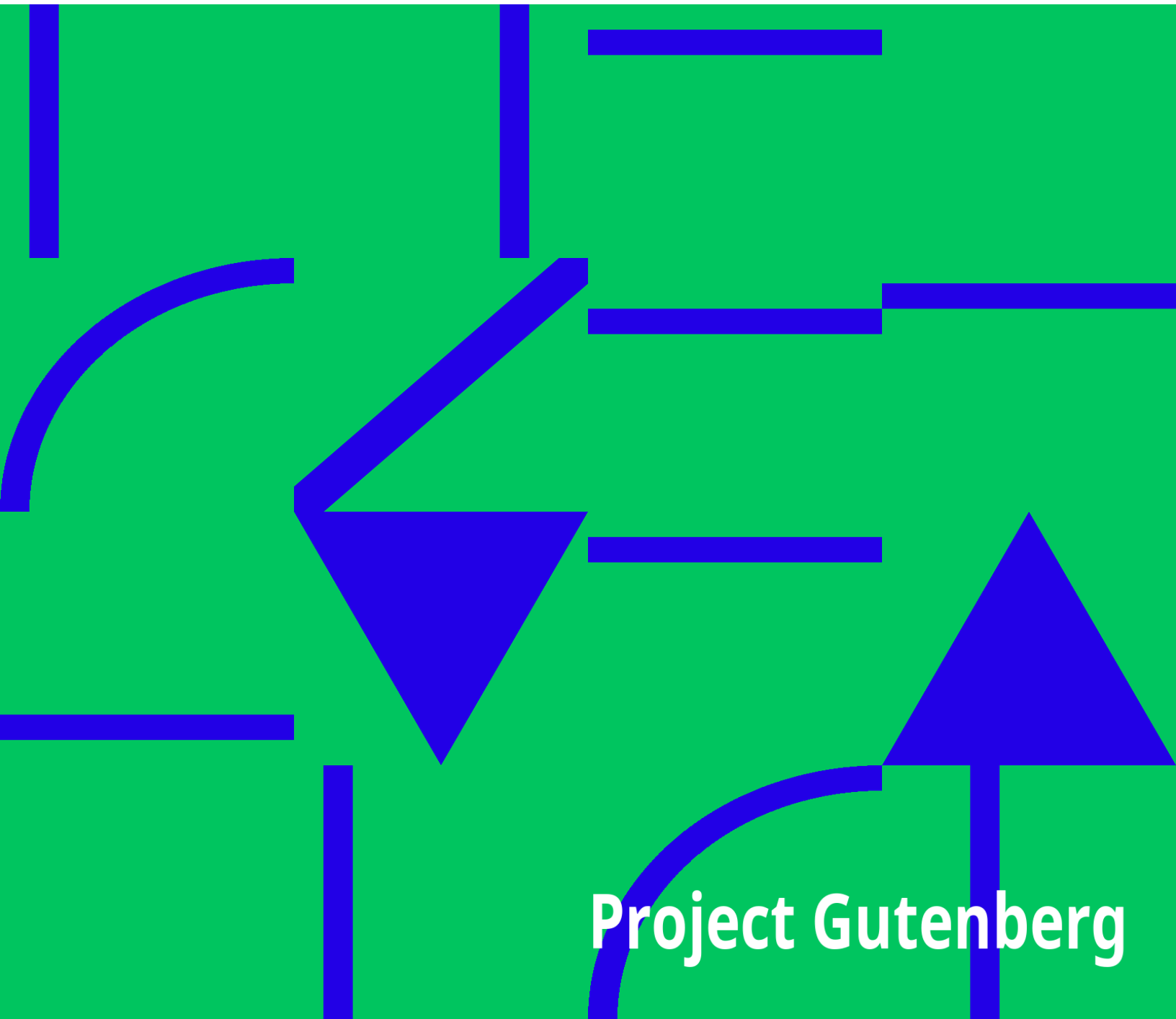


The Undersea Tube

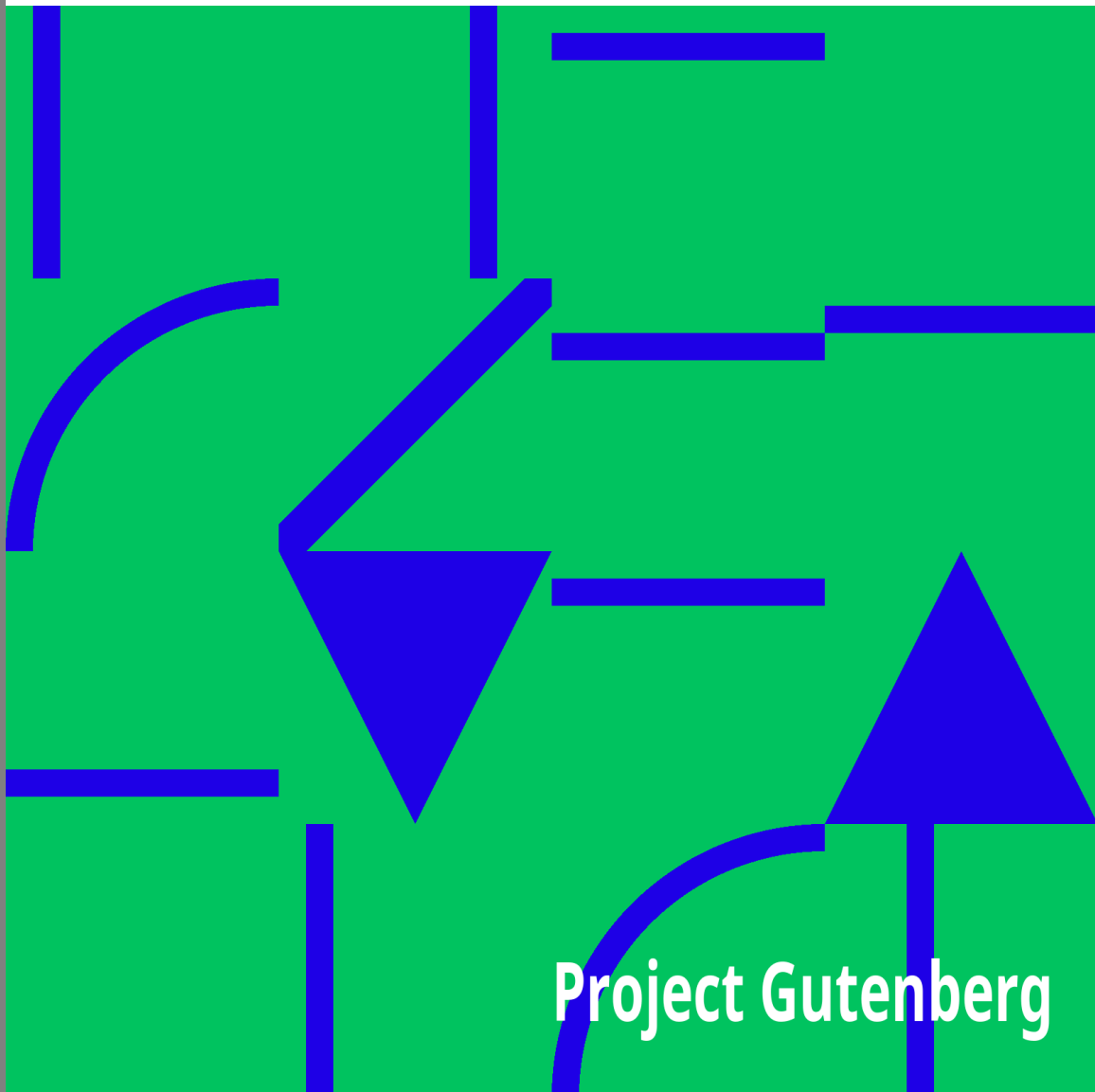
L. Taylor Hansen



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The Undersea Tube

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*** START OF THE PROJECT GUTENBERG EBOOK THE
UNDERSEA TUBE ***

The Undersea Tube

**BY L.
TAYLOR
HANSEN**

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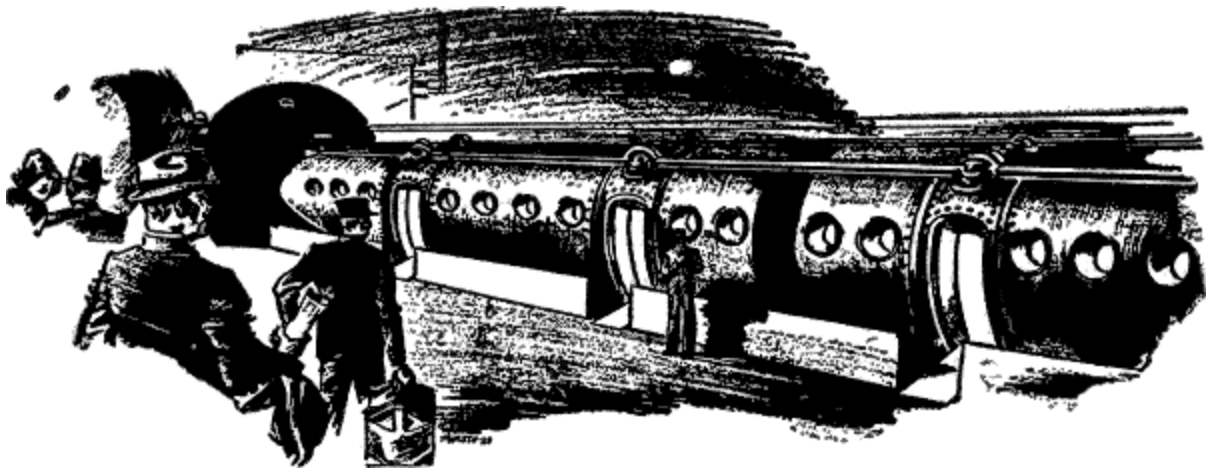
IF my friend the engineer had not told me the Tube was dangerous, I would not have bought a ticket on that fatal night, and the world would never have learned the story of the Golden Cavern and the City of the Dead. Having therefore, according to universal custom, first made my report as the sole survivor of the much-discussed Undersea Tube disaster to the International Committee for the Investigation of Disasters, I am now ready to outline that story for the world. Naturally I am aware of the many wild tales and rumors that have been circulated ever since the accident, but I must ask my readers to bear with me while I attempt to briefly sketch, not only the tremendous difficulties to be overcome by the engineers, but also the wind-propulsion theory which was made use of in this undertaking; because it is only by understanding something of these two phases of the Tube's engineering problems that one can understand the accident and its subsequent revelations.

It will be recalled by those who have not allowed their view of modern history to become too hazy, that the close of the twentieth century saw a dream of the engineering world at last realized—the completion of the long-heralded undersea railroad. It will also be recalled that the engineers in charge of this stupendous undertaking were greatly encouraged by the signal success of the first tube under the English Channel, joining England and France by rail. However, it was from the second tube across the Channel and the tube connecting Montreal to New York, as well as the one connecting New York and Chicago, that they obtained some of their then radical ideas concerning the use of wind power for propulsion. Therefore, before the Undersea Tube had been completed, the engineers in charge had decided to make use of the new method in the world's longest tunnel, and upon that decision work was immediately commenced upon the blue-prints for the great air pumps that were to rise at the two ends—Liverpool and New York. However, I will touch upon the theory of wind-propulsion later and after the manner in which it was explained to me.

It will be recalled that after great ceremonies, the Tube was begun simultaneously at the two terminating cities and proceeded through solid rock—low enough below the ocean floor to overcome the terrible pressure of the body of water over it, and yet close enough to the sea to overcome the intensity of subterranean heat. Needless to say, it was an extremely hazardous undertaking, despite the very careful surveys that had been made, for the little parties of workmen could never tell when they would strike a crack or an unexpected crevice that would let down upon them with a terrible rush, the waters of the Atlantic. But hazard is adventure, and as the two little groups of laborers dug toward each other, the eyes of the press followed them with more persistent interest than it has ever followed the daily toil of any man or group of men, either before or since.

ONCE the world was startled by the "extree-ee—" announcing that the English group had broken into an extinct volcano, whose upper end had apparently been sealed ages before, for it contained not water but air—curiously close and choking perhaps, but at least it was not the watery deluge of death. And then came the great discovery. No one who lived through that time will forget the thrill that quickened the pulse of mankind when the American group digging through a seam of old lava under what

scientists call the "ancient ridge," broke into a sealed cavern which gleamed in the probing flashlights of the workers like the scintillating points of a thousand diamonds. But when they found the jeweled casket, through whose glass top they peered curiously down upon the white body of a beautiful woman, partly draped in the ripples of her heavy, red hair, the world gasped and wondered. As every school child knows, the casket was opened by curious scientists, who flocked into the tube from the length of the world, but at the first exposure to the air, the strange liquid that had protected the body vanished, leaving in the casket not the white figure, but only a crumbling mass of grey dust. But the questions that the finding of the cave had raised remained unanswered.



Who was this woman? How did she get into the sealed cavern? If she had been the court favorite of that mythical kingdom, now sunk beneath the waves, and had been disposed of in court intrigue, why would her murderers have buried her in such a casket? How had she been killed? An unknown poison? Perhaps she had been a favorite slave of the monarch. This view gained many converts among the archaeologists who argued that from all the evidence we have available, the race carrying the Iberian or Proto-Egyptian culture, long thought to have been the true refugees from sinking Atlantis, were a slight dark-haired race. Therefore this woman must have been a captive. Geologists, analyzing the lava, announced that it had hardened in air and not in water, while anthropologists classed the skull of the woman as essentially more modern than either the Neanderthal or Cro-Magnon types. But the engineers, secretly fuming at the delay, finally managed to fill up the cave and press on with their drills.

Then following the arguments that still flourished in the press, came a tiny little news article and the first message to carry concern to the hearts of the engineers. The sea had begun to trickle in through one slight crack. Perhaps it was only because the crevice was located on the English side of the now famous "ancient ridge" that the article brought forth any notice at all. But for the engineers it meant the first warning of possibly ultimate disaster. They could not seal the crack, and pumps were brought into play. However, as a month wore on, the crack did not appear to widen to any material extent and the danger cry of a few pessimists was forgotten.

Finally, it will be remembered, that sounders listening in the rocks heard the drillers of the other party, and then with wild enthusiasm the work was pushed on to completion. The long Tube had been dug. Now it only remained for the sides at the junction to be enlarged and encased with cast iron, while the work of setting up the great machines designed to drive the pellet trains through, was also pushed on to its ultimate end. Man had essayed the greatest feat of engineering ever undertaken in the history of the planet, and had won. A period of wild celebration greeted the first human beings to cross each direction below the sea.

Did the volume of water increase that was carried daily out of the Tube and dumped from the two stations? If it did, the incident was ignored by the press. Instead, the fact that some "cranks" persisted in calling man's latest toy unsafe, only attracted more travel. The Undersea Tube functioned on regular schedule for three years, became the usual method of ocean transit.

THIS was the state of matters, when on the fourth of March last, our textile company ordered me to France to straighten out some orders with the France house, the situation being such that they preferred to send a man. Why they did not use radio-vision I do not care to state, as this is my company's business.

Therefore, upon entering my apartment, I was in the midst of packing when the television phone called me. The jovial features of "Dutch" Higgins, my one-time college room-mate and now one of the much-maligned engineers of the Undersea Tube, smiled back at me from the disk.

"Where are you? I thought we had a sort of dinner engagement at my apartment, Bob."

"By gollies I forgot, Dutch. I'll be right over—before it gets cold."

Then immediately I turned the knob to the Municipal Aerial-car yards, and ordered my motor, as I grabbed my hat and hurried to the roof. In due time, of course, I sprang the big surprise of the evening, adding:

"And, of course, I'm going by the Tube, I feel sort of a half-partnership in it because you were one of the designers."

A curious half-pained look crossed his face. We had finished our meal, and were smoking with pushed-back chairs. He finished filling his pipe, and scowled.

"Well? Why don't you say something? Thought you'd be—well, sort of pleased."

He struck his automatic lighter and drew in a long puff of smoke before answering.

"Wish you'd take another route, Bob."

"Take another route?"

"Yes. If you want it straight, the Tube is not safe."

"You are joking."

But as I looked into his cold, thoughtful blue eyes, I knew he had never been more serious.

"I wish that you would go by the Trans-Atlantic Air Liners. They are just as fast."

"But you used to be so enthusiastic about the Tube, Dutch! Why I remember when it was being drilled that you would call me up at all kinds of wild hours to tell me the latest bits of news."

He nodded slowly.

"Yes, that was in the days before the crack."

"Yet you expected to take care of possible leaks, you know," I countered.

"But this crack opened after the tunnel had been dug past it, and lately it has opened more."

"Are the other engineers alarmed?"

"No. We are easily taking care of the extra water and again the opening seems to remain at a stationary width as it has for the past three years. But we cannot caulk it."

"Are you going to publish these views?"

"No. I made out a minority report. I can do no more."

"Dutch, you are becoming over-cautious. First sign of old age."

"Perhaps," with the old smile.

"But after all it is now more than three years since we have had a talk on the Tube. After it began to function as well as the Air-Express you sort of lost interest in it."

"And the world did too."

"Certainly—but the public ever was a fickle mistress. Who said that before me?"

He laughed and blew out a long puff of smoke.

"Everyone, Bob."

"But as to the Tube, if I cross under the sea, I would want to be as well informed on the road as I was three years ago. Now in the meantime, you have dropped interest in the long tunnel while I have become more interested in textiles—with the result that I have forgotten all I ever did know—which compared to your grasp of the details, was little enough."

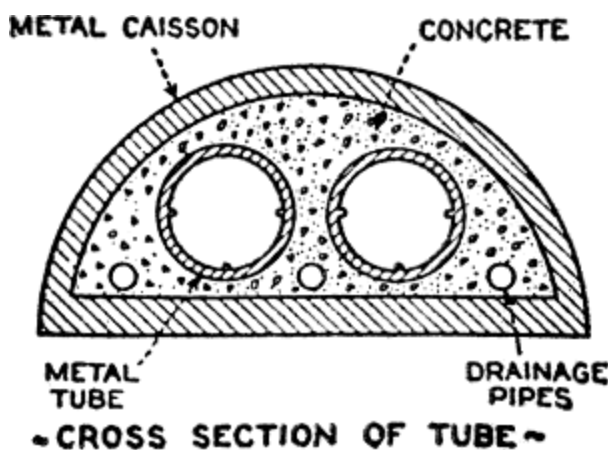
BUT his face showed none of the old-time animation on the subject. What a different man, I mused to myself, from that enthusiastic engineering student that I used to come upon dreaming over his blue-prints. He was considered "half-cracked" in those days when he would enthuse over his undersea railroad, but his animated face was lit with inspiration. Now the light was gone.

"Well, Dutch, how about it? Aren't you going to make me that brief little sketch of the length plan and cross-section of the Tube? I remember your sketch of it in college, and it tends to confuse me with the real changes that were made necessary when the wind-propulsion method was adopted."

"All right, old timer. You remember that the Tube was widened at the sides in order that we could make two circular tubes side by side—one going each way."

"I had forgotten that they were circular."

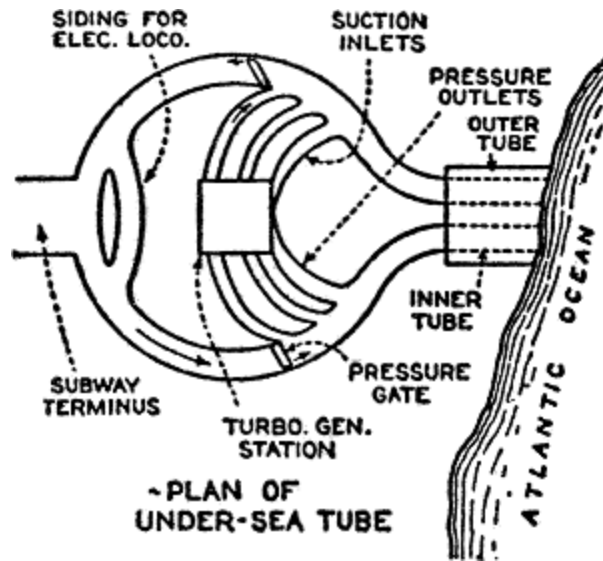
"That is because of the pressure. A circle presents the best resistance," and picking an odd envelope from his pocket, he made the following sketch and passed it to me.



I nodded as I recognized the cross-section.

"Now the plan of the thing is like this," he added, putting aside his pipe and pulling a sheet of paper from the corner of his desk.

Rapidly, with all his old accuracy, he sketched the main plan and leaned over as he handed it to me.



"You see," he explained, picking up his pipe again, "both pumps work at one time—in fact, I should say all four, because this plan is duplicated on the English side. On both ends then, a train is gently pushed in by an electric locomotive. A car at a time goes through the gate so that there is a cushion of air between each car. The same thing happens at Liverpool. Now, when the due train comes out of the suction tube, it goes on out the gate, but the air behind it travels right on around and comes in behind the train that is leaving."

"But how are you assured that it will not stall somewhere?"

"It won't be likely to with pressure pumps going behind it and suction pumps pulling from in front. We can always put extra power on if necessary. Thus far the road has worked perfectly."

"How much power do you need to send it through, under normal conditions?"

"Our trains have been averaging about fifty tons, and for that weight we have found that a pound pressure is quite sufficient. Now, taking the tunnel's length as four thousand miles (of course it is not that long, but round figures are most convenient) and the tube width eleven and one quarter feet each and working this out we have 3,020,000 cubic feet of free air per minute or 2,904,000 cubic feet of compressed air, which would use about 70,000 horse power on the air compressor."

"But isn't the speed rather dizzy?"

"Not any more dizzy, Bob, than those old fashioned money-carrying machines that the department stores used to use—that is in comparison to size. The average speed is about 360 feet a second. Of course, the train is allowed to slow down toward the end of its run, even before it hits the braking machinery beyond the gate."

"But how much pressure did you say would be put on the back of the diaphragm—I remember that each car has a flat disc on the back that fits fairly tightly to the tube ..."

"The pressure on the back is less than seven tons. However, the disc does not fit tight. There are several leaks. For instance, the cars are as you know, run on the principle of the monorail with a guiding rail on each side. The grooves for the rails with their three rollers are in each car. There is a slight leakage of air here."

"You used the turbo type of blower, didn't you?"

"Had to because of the noise. We put some silencing devices on that and yet we could not kill all of the racket. However a new invention has come up that we will make use of soon now."

"**B**UT I can't understand, Dutch, why you seemed so put out when I announced my intention of going to Europe via the Tube. Why, I can remember the day when that would have tickled you to death."

"You followed the digging of the Tube, didn't you?"

"Yes, of course."

"You remember the volcano and lava seams?"

"Yes."

"Well, I do not believe that the crack was a pressure crevice. If it had been, we were far enough below the ocean floor to have partly relieved the situation by the unusually solid building of the Tube. The tremendous shell of this new type of specially hardened metal—"

"And the rich concrete that was used as filling! That was one job no one slipped up on. I remember how you watched it—"

"Yet the crack has widened, Bob, since the Tube was completed."

"How can you be certain?"

"By the amount of water coming through the drain pipes."

"But you said that once more it was stationary."

"Yes, and that is the very thing that proves, I believe, the nature of the crack."

"I don't follow you."

"Why it isn't a crack at all, Bob. It is an earthquake fault."

"Good heavens, you don't mean—"

"Yes, I do. I mean that the next time the land slips our little tube will be twisted up like a piece of string, or crushed like an eggshell. That always was a rocky bit of land. I thought in going that far north, though, that we had missed the main line of activity; I mean the disturbances that had once wiped out a whole nation, if your scientists are correct."

"Then you mean that it is only a matter of time?"

"Yes, and I have been informed by one expert that the old volcanic activity is not dead either."

"So that is what has stolen away your laugh?"

"Well I am one of the engineers—and they won't suspend the service."

"Fate has played an ugly trick on you, Dutch, and through your own dreams too. However, you have made me decide to go by the Tube."

He took his pipe out of his mouth and stared at me.

"Sooner or later the Tube will be through, and I have never been across. Nothing risked—a dull life. Mine has been altogether too dull. I am now most certainly going by the Tube."

A bit of the old fire lit up his eyes.

"Same old Bob," he grunted as I rose, and then he grasped my hand with a grin.

"Good luck, my boy, on your journey, and may old Vulcan be out on a vacation when you pass his door."

Thus we said good-by. I did not know then that I would never see him again—that he also took the train that night in order to make one last plea to the International Committee, and so laid down his life with the passengers for whom he had pleaded.

It was with many conflicting thoughts, however, that I hurried to the great Terminus that fatal night, where after being ticketed, photographed and tabulated by an efficient army of clerks, I found myself in due time, being ushered to my car of the train.

FOR the benefit of those who have never ridden upon the famous "Flier," I could describe the cars no better than to say that coming upon them by night as I did, they looked like a gigantic, shiny worm, of strange shape, through whose tiny port-holes of heavy glass in the sides, glowed its luminous vitals.

I was pompously shown to the front car, which very much resembled a tremendous cartridge—as did all of the other segments of this great glow-worm.

Having dismissed the porter with a tip and the suspicion that my having the front car was the work of my friend, who was willing to give me my money's worth of thrill, and that the porter was aware of this, I stowed away my bags and started to get ready for bed. I had no sooner taken off my coat than the door was opened and an old fellow with a mass of silver hair peered in at me.

"I beg your pardon, sir, but I understand you have engaged this car alone?"

"Yes."

"I can get no other accommodations tonight. You have an extra berth here and I must get to Paris tomorrow. I will pay you well—"

I smiled.

"Take it. I was beginning to feel lonesome, anyway."

He bowed gravely and ordered the porter to bring in his things. I decided he was a musician. Only artists go in for such lovely hair. But he undressed in dignified silence, not casting so much as another glance in my direction, while on my part I also forgot his presence when, looking through the port-hole, I realized that the train had begun to move. Soon the drone of the propelling engines began to make itself heard. Then the train began to dip down and the steel sides of the entrance became too high for me to see over. My friend of the silver hair had already turned off the light, and now I knew by the darkness that we had entered the Tube. For some time I lay awake thinking of "Dutch" and the ultimate failure of his life's dream, as he had outlined it to me, and then I sank into a deep, dreamless sleep.

I was awakened by a terrible shock that hurled me up against the side of the compartment. A dull, red glow poured through the port-hole, lighting up the interior with a weird, bloody reflection. I crept painfully up to the port-hole and looked out. The strangest sight that man has ever looked upon met my eyes. The side of the wall had blown out into a gigantic cavern, and with it the rest of the cars had rolled down the bluff a tangled, twisted mass of steel. My car had almost passed by, and now it still stuck in the tube, even though the last port-hole through which I peered seemed to be suspended in air. But it was not the wrecked cars from which rose such wails of despair and agony that held my attention, but the cavern itself. For it was not really a cave, but a vast underground city whose wide, marble streets stretched away to an inferno of flame and lava. By the terrible light was lit up a great white palace with its gold-tipped scrolls, and closer to me, the golden temple of the Sun, with its tiers of lustrous yellow stairs—stairs worn by the feet of many generations.

Above the stairs towered the great statue of a man on horseback. He was dressed in a sort of tunic, and in his uplifted arm he carried a scroll as if for the people to read. His face was turned toward me, and I marveled even in that wild moment that the unknown sculptor could have caught such an

expression of appeal. I can see the high intellectual brow as if it were before me at this moment—the level, sympathetic eyes and the firm chin.

THEN something moving caught my eyes, and I swear I saw a child—a living child coming from the burning city—running madly, breathlessly from a wave of glowing lava that threatened to engulf him at any moment. In spite of all the ridicule that has been showered upon me, I still declare that the child did not come from the wreckage and that he wore a tunic similar to the one of the statue and not the torn bit of a nightgown or sheet.

He was some distance from me, but I could plainly see his expression of wild distraction as he began to climb those gleaming stairs. Strangely lustrous in the weird light, was that worn stairway of gold—gold, the ancient metal of the Sun. With the slowness of one about to faint he dragged himself up, while his breath seemed to be torn from his throat in agonizing gasps. Behind him, the glowing liquid splashed against the steps and the yellow metal of the Sun began to drip into its fiery cauldron.

The child reached the leg of the horse and clung there.

... Then suddenly the whole scene began to shake as if I had been looking at a mirage, while just behind my car I had a flashing glimpse in that lurid light of an emerald-green deluge bursting in like a dark sky of solid water, and in that split-second before a crushing blow upon my back, even through that tangle of bedclothes, knocked me into unconsciousness, I seemed to hear again the hopeless note in the voice of my friend as he said:

"—an earthquake fault."

After what seemed to me aeons of strange, buzzing noises and peculiar lights, I at last made out the objects around me as those of a hospital. Men with serious faces were watching me. I have since been told that I babbled incoherently about "saving the little fellow" and other equally incomprehensible murmurings. From them I learned that the train the other way was washed out, a tangled mass of wreckage just like my car, both terminus stations wrecked utterly, and no one found alive except myself. So, although I am to be a hopeless cripple, yet I am not sorry that the skill

and untiring patience of the great English surgeon, Dr. Thompson, managed to nurse back the feeble spark of my life through all those weeks that I hung on the borderland; for if he had not, the world never would have known.

As it is, I wonder over the events of that night as if it had not been an experience at all—but a wild weird dream. Even the gentleman with the mass of silver hair is a mystery, for he was never identified, and yet in my mind's recesses I can still hear his cultured voice asking about the extra berth, and mentioning his pressing mission to Paris. And somehow, he gives the last touch of strangeness to the events of that fatal night, and in my mind, he becomes a part of it no less than the child on the stairs, the burning inferno that lit the background, and the great statue of that unknown hero who held out his scroll for a moment in that lurid light, like a symbol from the sunken City of the Dead.

THE END

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